LP5 Detailed Viva Questions & Answers (HPC + Deep Learning)

This PDF contains the most important and frequently asked LP5 viva questions with descriptive answers. The answers are written in a simple and effective viva style according to the SPPU LP5 pattern.

SECTION 1 : HIGH PERFORMANCE COMPUTING (HPC)

Q1. What is High Performance Computing (HPC)?

High Performance Computing (HPC) is the use of multiple processors or computers working together to solve complex computational problems quickly. It increases execution speed and improves performance for large-scale applications. HPC is widely used in scientific simulations, AI/ML, weather forecasting, and research.

Q2. What is OpenMP?

OpenMP stands for Open Multi-Processing. It is an API used for parallel programming in C, C++, and Fortran on shared memory systems. OpenMP uses compiler directives like #pragma omp parallel to create multiple threads and execute tasks simultaneously.

Q3. Difference between sequential and parallel algorithm?

A sequential algorithm executes one instruction after another using a single processor. A parallel algorithm divides the work among multiple processors or threads and executes tasks simultaneously. Parallel algorithms reduce execution time and improve efficiency for large computations.

Q4. What is Parallel BFS?

Parallel Breadth First Search explores graph nodes level by level using multiple threads. Different vertices at the same level can be processed simultaneously, which improves performance. OpenMP tasks or parallel loops are commonly used to implement parallel BFS.

Q5. What is Parallel DFS?

Parallel DFS explores different branches or subtrees of a graph simultaneously using multiple processors. It improves execution speed for large graphs. Dynamic load balancing is important because different branches may contain different amounts of work.

Q6. Why is dynamic load balancing important in DFS?

In DFS, some processors may finish earlier because different subtrees have different sizes. Dynamic load balancing redistributes work among processors to avoid idle processors. This improves processor utilization and overall performance.

Q7. What is reduction in OpenMP?

Reduction is an OpenMP technique used to combine partial results from multiple threads into a single final result. It is commonly used for operations like sum, minimum, maximum, and average. The reduction clause prevents race conditions during parallel execution.

Q8. Why is Bubble Sort difficult to parallelize?

Bubble sort contains data dependencies because swapping one pair of elements affects neighboring comparisons. Due to this dependency, it is difficult to execute completely in parallel. Odd-even transposition sort is used as a parallel variation of bubble sort.

Q9. What is Odd-Even Transposition Sort?

Odd-even transposition sort is a parallel version of bubble sort. In one phase, odd indexed elements are compared and swapped, and in the next phase, even indexed elements are processed. This allows multiple comparisons to occur simultaneously.

Q10. What is Merge Sort and why is it suitable for parallelization?

Merge sort is a divide-and-conquer algorithm that recursively divides the array into smaller subarrays and merges them after sorting. Since subarrays are independent, they can be processed by different threads simultaneously. Therefore, merge sort is highly suitable for parallel programming.

Q11. What is CUDA?

CUDA stands for Compute Unified Device Architecture developed by NVIDIA. It allows developers to use GPU power for parallel computing. CUDA is widely used for applications like matrix multiplication, AI training, and image processing.

Q12. What is a CUDA Kernel?

A CUDA kernel is a function that runs on the GPU and executes in parallel by many threads. Each thread performs a small part of the computation. Kernels are launched using grid and block configurations.

Q13. Difference between CPU and GPU?

A CPU contains fewer powerful cores optimized for sequential tasks and general-purpose computing. A GPU contains thousands of smaller cores optimized for parallel execution. GPUs are better for repetitive and data-intensive tasks like graphics and deep learning.

Q14. What is Speedup in Parallel Computing?

Speedup measures the performance improvement achieved using parallel execution. It is calculated as the ratio of sequential execution time to parallel execution time. Higher speedup indicates better parallel performance.

Q15. Difference between OpenMP and MPI?

OpenMP is used for shared memory parallel programming within a single system using threads. MPI is used for distributed memory systems where processes communicate using message passing. OpenMP is easier for multicore systems, while MPI is suitable for clusters.

SECTION 2 : DEEP LEARNING (DL)

Q1. What is Deep Learning?

Deep Learning is a subset of machine learning that uses artificial neural networks with multiple hidden layers. It automatically learns features from large amounts of data. Deep learning is widely used in image recognition, speech processing, NLP, and recommendation systems.

Q2. What is a Deep Neural Network (DNN)?

A Deep Neural Network consists of an input layer, multiple hidden layers, and an output layer. Each layer processes information and passes it to the next layer. DNNs are capable of learning complex patterns from data and improving prediction accuracy.

Q3. What is Linear Regression?

Linear regression is a supervised learning algorithm used to predict continuous numerical values. It establishes a linear relationship between input variables and output variables. In LP5, it is used for Boston House Price prediction.

Q4. Why is standardization important in Deep Learning?

Standardization scales data to zero mean and unit variance so that all features contribute equally during training. It improves convergence speed and model accuracy. Standardization also prevents features with larger values from dominating the learning process.

Q5. Why do we split data into training and testing sets?

Training data is used to train the model and learn patterns from the dataset. Testing data is used to evaluate the model performance on unseen data. This helps in checking whether the model generalizes properly or suffers from overfitting.

Q6. What is an Epoch?

An epoch means one complete pass of the entire training dataset through the neural network. During each epoch, model weights are updated to minimize error. Increasing epochs may improve accuracy but too many epochs can cause overfitting.

Q7. What is ReLU Activation Function?

ReLU stands for Rectified Linear Unit and is one of the most commonly used activation functions in deep learning. It returns zero for negative inputs and returns the same value for positive inputs. ReLU helps reduce the vanishing gradient problem and speeds up training.

Q8. What is Binary Classification?

Binary classification is a supervised learning task where data is classified into two categories such as positive-negative or yes-no. In LP5, IMDB movie review sentiment analysis is an example of binary classification. The output is usually represented as 0 or 1.

Q9. What is Binary Cross Entropy?

Binary cross entropy is a loss function used for binary classification problems. It measures the difference between predicted probabilities and actual class labels. Lower loss values indicate better model performance.

Q10. What is Adam Optimizer?

Adam optimizer is an advanced optimization algorithm used to update weights during neural network training. It combines momentum and adaptive learning rate techniques. Adam is widely used because it provides faster convergence and better performance.

Q11. What is CNN?

CNN stands for Convolutional Neural Network and is mainly used for image processing tasks. CNN automatically extracts important features such as edges, textures, and patterns using convolution layers. It is widely used in image classification and object detection.

Q12. Why is CNN suitable for image classification?

CNN can automatically learn spatial features from images without manual feature extraction. It uses convolution and pooling operations to reduce complexity while preserving important information. Therefore, CNN achieves high accuracy in image-related tasks.

Q13. What is Pooling in CNN?

Pooling is a technique used to reduce the dimensions of feature maps in CNN. It decreases computational complexity and helps prevent overfitting. Max pooling is the most common pooling operation used in CNN models.

Q14. What is RNN?

RNN stands for Recurrent Neural Network and is designed for sequential or time-series data. It stores previous information using recurrent connections, making it useful for language modeling and stock prediction. RNNs are commonly used in NLP and forecasting tasks.

Q15. What is LSTM?

LSTM stands for Long Short-Term Memory and is a special type of RNN. It solves the vanishing gradient problem and can remember long-term dependencies. LSTM is widely used for time-series forecasting and speech recognition.

Final Viva Tips

- Explain every algorithm step-by-step instead of only definitions.
- Mention advantages of parallel computing whenever possible.
- Remember OpenMP directives like parallel, for, task, reduction, critical, and atomic.
- For DL viva, explain the working flow: dataset → preprocessing → model → training → prediction.
- Know practical datasets like Boston Housing, IMDB, MNIST Fashion, and Google Stock Dataset.
- Be confident while explaining CNN, DNN, and RNN architecture.